

work) provides a direct interface for previewing contents of files that your app does not support directly.

Safari Services framework. Safari Services framework (SafariServices.framework) provides support for programmatically adding URLs to the user's Safari reading list.

Social framework. Social framework (Social.framework) provides a simple interface for accessing the user's social media accounts.

StoreKit framework. StoreKit framework (StoreKit.framework) provides support for purchasing content and services from within your iOS apps, a feature known as In-App Purchase.

System Configuration framework. System Configuration framework (SystemConfiguration.framework) provides reachability interfaces, which you can use to determine the network configuration of a device.

WebKit framework. WebKit framework (WebKit.framework) lets you display HTML content in your app.

Core OS layer. Core OS layer contains low-level features that most other technologies are built upon. Even if you do not use these technologies directly in your apps, these are most likely being used by other frameworks. And in situations where you need to explicitly deal with security or communicating with an external hardware accessory, you do so using the frameworks in this layer.

Accelerate framework. Accelerate framework (Accelerate.framework) contains interfaces for performing digital signal processing (DSP), linear algebra and image-processing calculations.

Core Bluetooth framework. Core Bluetooth framework (CoreBluetooth.framework) allows developers to interact specifically with Bluetooth Low Energy (BLE) accessories.

External Accessory framework. External Accessory framework (ExternalAccessory.framework) provides support for communicating with hardware accessories attached to an

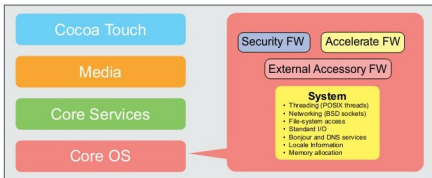


Fig. 7: Core OS layer

iOS based device.

Generic Security Services framework. Generic Security Services framework (GSS.framework) provides a standard set of security-related services to iOS apps.

Local Authentication framework. Local Authentication framework (LocalAuthentication.framework) lets you use Touch ID to authenticate a user.

Network Extension framework. Network Extension framework (NetworkExtension.framework) provides support for configuring and controlling Virtual Private Network (VPN) tunnels.

Security framework. In addition to its built-in security features, iOS also provides an explicit Security framework (Security.framework) that you can use to guarantee the security of the data your app manages.

System. The system level encompasses the kernel environment, drivers and low-level UNIX interfaces of the OS.

64-bit support for iOS

iOS was initially designed to support binary files on devices using a 32-bit architecture. In iOS 7, however, support was introduced for compiling, linking and debugging binaries on a 64-bit architecture. All system libraries and frameworks are 64-bit ready, meaning that these can be used in both 32-bit and 64-bit apps. When compiled for 64-bit runtime, apps may run faster because of the availability of extra processor resources in 64-bit mode.

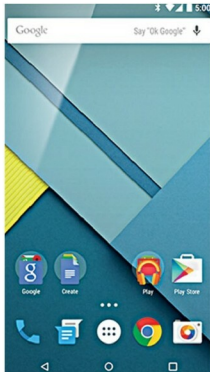


Fig. 8: Android 5.0 Home screen

Android

Android is a mobile OS based on Linux kernel and currently developed by Google. With a UI based on direct manipulation, Android is designed primarily for touchscreen mobile devices such as smartphones and tablet computers, with specialised UIs for televisions (Android TV), cars (Android Auto) and wrist watches (Android Wear). Android Inc. was founded in Palo Alto, California, the USA, in October 2003.

Architecture. Android is architected in the form of a software stack comprising applications, an OS, runtime environment, middleware, ser-